

Quantitative computerized analysis demonstrates strongly compartmentalized tissue deformation patterns underlying mammalian heart tube formation

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This **fundamental** work substantially advances our understanding of tissue deformation and growth patterns during the earliest stages of mammalian heart development. One of the strengths of the work is the **compelling** quantitative approach to analyzing time-lapse imaging data using an original computational pipeline, which goes beyond the current state of the art and provides new insights into heart tube formation. Overall, this rigorous study will be of broad interest to computational and developmental biologists studying tissue dynamics.

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Abstract

Summary

The quantitative analysis of tissue deformation at cellular resolution remains an important challenge in mammalian organogenesis. Here, we developed a new computational framework to extract regional and temporal patterns of tissue deformation and applied it to a collection of live microscopy datasets from mouse cardiogenesis. We devised a method to track tissue deformation directly from time-lapse raw images and experimentally validated the method by comparison with actual cell tracks. We then used a machine-learning approach to temporally and spatially align different specimens and reconstruct a single statistical model of tissue motion, deducing maps of strain, anisotropy, and tissue growth. We also implemented a virtual fate mapping tool that allows tracking any initial position in the cardiac primordium onto the linear heart tube. Our study reveals predominant local cellular coherence during the deformation of the cardiac tissue, whereas strong compartmentalization of tissue deformation patterns transforms the bilateral cardiac

primordium into a 3D longitudinal heart tube. At the future outer curvature of the primitive tube, the ventricular chamber forms by expansion of the tissue in a hemi-barrel shape with two harnessing belts; one that constrains tissue expansion at the arterial pole and one that constrains the expansion at the venous pole. Our study provides a new approach to understanding organ morphogenesis and proposes a new model of primitive heart tube formation.

Introduction

Quantifying tissue morphogenesis at cellular level is essential for understanding developmental mechanisms, addressing congenital heart disease, and designing tissue engineering strategies. In mammals, the heart is the first organ to form and function. Heart morphogenesis is particularly challenging to study because it involves extreme tissue deformation and occurs concurrently with the onset of cardiac beating. Early cardiogenesis involves extensive reorganization and progressive differentiation of the cardiogenic fields to rapidly create a pumping structure (Ivanovitch et al., 2017 [↗](#); Kelly et al., 2014 [↗](#); Meilhac et al., 2014 [↗](#); Buckingham et al., 2005 [↗](#); Moorman & Christoffels, 2003 [↗](#)). In mouse, the first differentiation wave produces a simple single layer known as the cardiac crescent (CC). The CC then undergoes a series of deep deformations, gradually forming the heart tube (HT), a sophisticated three-dimensional structure with inflow (IFT) and outflow (OFT) tracts and ability to support embryonic circulation (de Boer et al., 2012 [↗](#)). While the evolution of tissue shape during cardiogenesis has been characterized and quantitatively analyzed (Ebrahimi et al., 2022 [↗](#); Esteban et al., 2022 [↗](#); Kawahira et al., 2020 [↗](#); J.-F. Le Garrec et al., 2013 [↗](#)), understanding how tissues deform during morphogenesis requires the definition of cell displacement and rearrangement patterns.

Recent advances in microscopy and genetic cell labelling have opened new avenues for investigating tissue dynamics. Live imaging techniques, enabling non-invasive observation of biological phenomena, have opened new opportunities and perspectives in this field (Dominguez et al., 2023 [↗](#); Raiola et al., 2023 [↗](#); Sendra et al., 2022 [↗](#); Ivanovitch et al., 2017 [↗](#); J. F. Le Garrec et al., 2017 [↗](#)). These new approaches have provided access to datasets that include tissue deformation and cellular tracks; however, these datasets remain underexploited for global, multiscale analysis of tissue deformation during cardiogenesis.

While some studies have focused on the dynamic analysis of discrete morphogenetic events of early heart development (Sendra et al., 2025 [↗](#); Ivanovitch et al., 2017 [↗](#); Buckingham et al., 2005 [↗](#); Meilhac et al., 2004 [↗](#)), the global dynamic analysis of tissue and cell dynamics during mammalian heart development has not been achieved. A thorough understanding of the morphogenetic mechanisms of early heart development requires adopting a multiscale approach that incorporates the behavior of individual cells, but at the same time describes the global dynamics at the tissue level.

Here we address this challenge by implementing a systematic and versatile methodology to quantify and compare tissue deformation patterns through integrating motion profiles estimated from multiple 3D live fluorescent microscopy images. The methodology used focuses on quantifying motion kinematics to compute deformation and the main strain directions, without considering the forces causing such motion. This is achieved through a non-invasive analysis made directly on individual live images. To align the individual motion profiles of multiple specimens over time, the framework incorporates a staging system that synchronizes 3D live images with a previously described pseudo-dynamic Atlas of tissue geometry during heart morphogenesis (Esteban et al., 2022 [↗](#)). This strategy allows to deduce deformation patterns with spatial and temporal resolution and including both instantaneous features and their cumulative history. The study performed showed essential features of cardiogenic tissue deformation indicating strong spatio-temporal compartmentalization of cell behaviors. The 3D+t model

generated also enables *in-silico* fate map analysis at the cellular or regional level, which provides a tool to interrogate specific features of the dynamics of cardiac cells during HT morphogenesis. This model lays the groundwork for future predictive simulations of the forces driving heart morphogenesis.

Results

Estimating Myocardial Motion and Deformation from Multiple Time-Lapse Datasets

We developed a workflow that consists of four steps summarized in **Figure 1**: (A) Individual time-lapse analysis to capture the geometry of the myocardium from the cardiac crescent to the linear heart tube; (B) Integration of multiple time-lapse datasets through spatiotemporal registration into a previously described 3D+t Atlas (Esteban et al., 2022); (C) Quantification of cardiac spatiotemporal deformation patterns during CC to HT morphogenesis; (D) *In-silico* fate mapping to investigate how different CC regions contribute to the forming HT. The full methodology is detailed in a co-submitted manuscript (Raiola et al., 2025b).

We analyzed both published (Ivanovitch et al., 2017) and new datasets consisting of 3D+t live images (Raiola et al., 2025b), including 16 embryos from early CC formation to the onset of heart looping. Myocardial cells were labelled using various reporter strategies: Nkx2.5GFP (Wu et al., 2006) or a combination of Nkx2.5Cre (Stanley et al., 2004) and various fluorescent reporter alleles. To enable individual cell tracking, Nkx2.5GFP embryos carried sparse genetic labels induced by low-dose tamoxifen using the RERT allele (CreERT2 knocked-in under the control of RNAPol2 promoter Guerra et al., 2003). Additional embryos expressed Mesp1Cre to mark all mesodermal cells, or Islet1Cre to specifically label the second heart field (SHF). Acquisitions were performed at a variety of temporal windows and with variable time-lapse periods. The procedures developed therefore cope with a diversity of labelling strategies, temporal rates and embryonic periods of acquisition (Raiola et al., 2025b).

To estimate tissue motion, we adapted the Medical Image Registration Toolbox (MIRT) (Myronenko & Song, 2010), which calculates displacement tensors across time on raw images (Figure 1A) (Raiola et al., 2025b; Myronenko & Song, 2010). This algorithm produces Tensor vectors (T) that describe the displacement of each image point along time (Figure 1A, orange grid). To validate these computational predictions, we compared them to manual tracking of >9000 genetically labelled cells through consecutive frames (Raiola et al., 2025b). The predicted displacements closely matched real cell movements even over extended imaging periods, validating the use of MIRT for high-resolution analysis of tissue deformation during HT formation.

To generate a unified model of cardiac tissue motion, we registered each time-lapse sequence to the 3D+t Atlas of heart development (Esteban et al., 2022). First, we segmented the myocardial tissue in each specimen of the time-lapse datasets and produced a dense mesh of triangles that describes the tissue surface (Raiola et al., 2025b). These meshes describe the dynamics of myocardial motion and were named “Live-Shapes”. Second, we developed a staging system based on a morphometric feature that aligns live images of different specimens over time and allows assignment of each specific specimen and stage to the corresponding Atlas geometry (Figure 1B) (Raiola et al., 2025b). Third, we designed a procedure to address the shape variability between equally staged hearts by mapping specimens onto the Atlas (Figure 1B) (Raiola et al., 2025b).

These procedures allowed us to map different parameters, such as strain and growth, from individual specimens onto the Atlas, providing their distribution and regional variability among specimens (Figure 1C, example of tissue growth mapping). Finally, using the tissue deformation

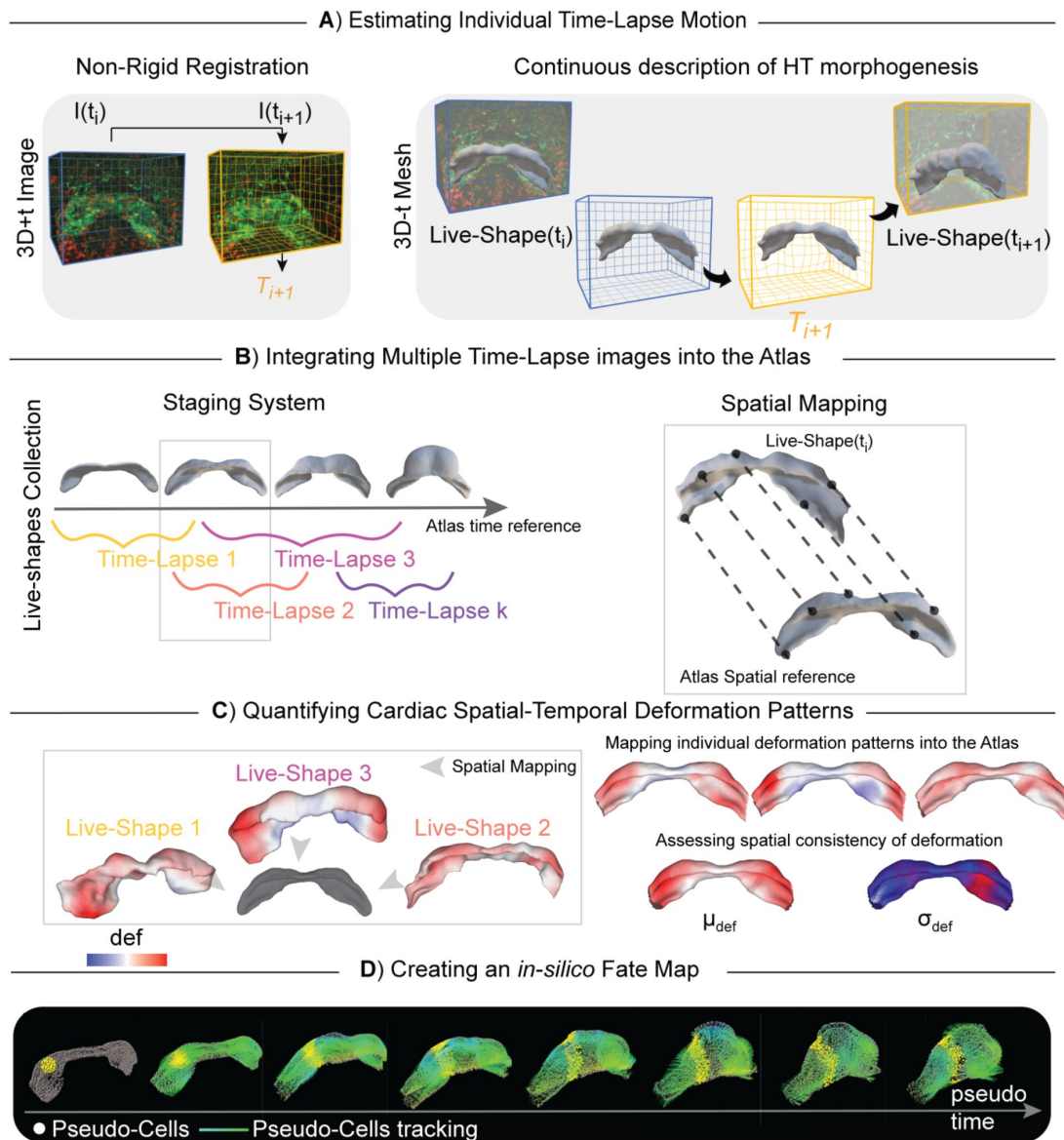


Figure 1

Workflow for extracting and defining the motion and deformation pattern of the myocardium during early heart morphogenesis.

The analysis pipeline consists of four main steps. Our dataset includes multiple specimens ranging from E7.75 to E8.25 (12 hours). A), Estimating Individual Time-Lapse Motion: A non-rigid algorithm extracts the transformation T needed to deform one frame, $I(t_i)$, into the next frame, $I(t_{i+1})$. The transformation is computed for a set of control points of a regular 3D grid (blue grid). Control point displacement is updated (orange grid) until the two frames overlap. We collect the set of transformations $\{T_i\}$ for each time-lapse image to compute the continuous description of HT morphogenesis. B), Integrating multiple time-lapse images into the Atlas: We incorporate the individual continuous descriptions of HT deformation into the Atlas. We define a staging system that aligns the individual time-lapse images on a common time reference and registers the related 3D mesh (Live-Shape) onto the spatial reference (Atlas). C), Quantifying Cardiac Spatial-Temporal Deformation Patterns: We extract the deformation patterns from each continuous description of HT deformation and compile them into the Atlas, generating a single stage-by-stage model (μ_{def} , σ_{def}). D), Creating an *in-silico* Fate Map: We concatenate a continuous description of HT deformation from the individual steps to recreate a unique and continuous description of early HT deformation patterns. Gray spots represent the initial positions of the pseudo-cells, while color lines, their displacement during development. Yellow spots indicate the tracked cells.

data, we performed *in-silico* fate mapping by simulating the forward displacement of pseudo-cells placed in the CC domain. This analysis showed how different CC regions contribute to specific domains of the forming HT and how they deform during this process (**Figure 1D** [↗](#)).

Extracting Deformation Patterns from the Continuous Description of Heart Tube Morphogenesis

To analyze tissue deformation patterns in the collection of Live-Shapes, we applied the continuous mechanics laws to the deformation of the mesh triangles (**Figure 2A** [↗](#), (Raiola et al., 2025b [↗](#)). Deformation was calculated between consecutive time points for 12 embryos, each with at least two shapes classified in successive Atlas stages. The resulting pattern was mapped onto the following stage shape, capturing the immediate deformation history. For each triangle, we computed local shape changes and applied smoothing (Raiola et al 2025b [↗](#)) to estimate local tissue growth rate (J), tissue anisotropy (θ) and the magnitude and direction tissue deformation (ϵ) (**Figure 2B** [↗](#)). In addition, we introduced a new parameter; “Strain Agreement Index” (ϕ), which quantifies local coordination of strain directions. This metric distinguishes regions of coordinated deformation (ordered) from areas with locally discrepant deformation directions (chaotic). It would also identify borders between regions with coherent but discrepant strain directions (**Figure 2B** [↗](#), STAR Methods – Strain Agreement Index). All deformation measurements are invariant to translation and rotation, making them robust to embryo drift during acquisition.

We then examined myocardial deformation pattern in two complementary ways: stepwise, by comparing deformation between consecutive stages; and cumulatively, by integrating the stepwise deformations through several stages (**Figure 2C** [↗](#)). To specifically focus on heart tube morphogenesis driven by tissue deformation, we excluded Atlas stage 1, given that between stages 1 and 2 substantial cell addition to the heart tube through myocardial differentiation takes place (Esteban et al., 2022 [↗](#)).

Stepwise Deformation Pattern Analysis

To investigate the spatial consistency of the deformation patterns among specimens, we mapped each deformation value from the Live-Shapes onto the corresponding Atlas geometry. For each stage group, we computed the spatial distributions of the mean deformation (μ_{def}) and its variability (σ_{def}) (**Figures S1** [↗](#)-**S4** [↗](#)). The stage2-to-stage3 and stage3-to-stage4 deformation analyses showed an inverse distribution of high growth and high anisotropy regions. High growth was associated with low anisotropy and high anisotropy appeared in regions with low growth. High anisotropy/low growth was mainly detected in a coherent region spanning the ventral midline region of the HT and extending laterally to the ventrocaudal region adjacent to the peripheral splanchnic mesoderm (see solid arrowheads in **Figure 2D** [↗](#), and asterisks in **Figure S1** [↗](#), Growth and Anisotropy columns). A second high anisotropy/low growth region appeared at the dorsal lips of the HT (see solid arrowheads in **Figure S1** [↗](#), Growth and Anisotropy columns). The rest of the HT, including the IFTs, showed a largely isotropic growth (**Figure S1** [↗](#)). In stage 4, a left-biased increase in growth was detected (**Figure S1** [↗](#), Growth column). Most specimens at this stage presented a coherent deformation direction, with strain oriented along the craniocaudal axis. Accordingly, they generally showed a high Agreement Index, indicating coordinated deformation of the myocardium. An exception to this is found at the lateral parts of the IFTs, where narrow bands of local strain disagreement were found separating the medial from the lateral parts of the IFTs (see empty arrowheads in the Agreement column). A closer observation of the strain alignment in these regions (see magnifications in the Max Strain column of **Figure S1** [↗](#)) shows that the low-agreement bands correspond to an abrupt transition between regions of coherent but discrepant strain directions. Whereas in the lateral-most regions of the IFTs strain aligns parallel to the main IFT axis, in their proximal region strain aligns craniocaudally, coherently with the rest of the HT (**Figure S1** [↗](#), Max Strain and Agreement columns). These results indicate a strong compartmentalization of collective cell behavior during HT

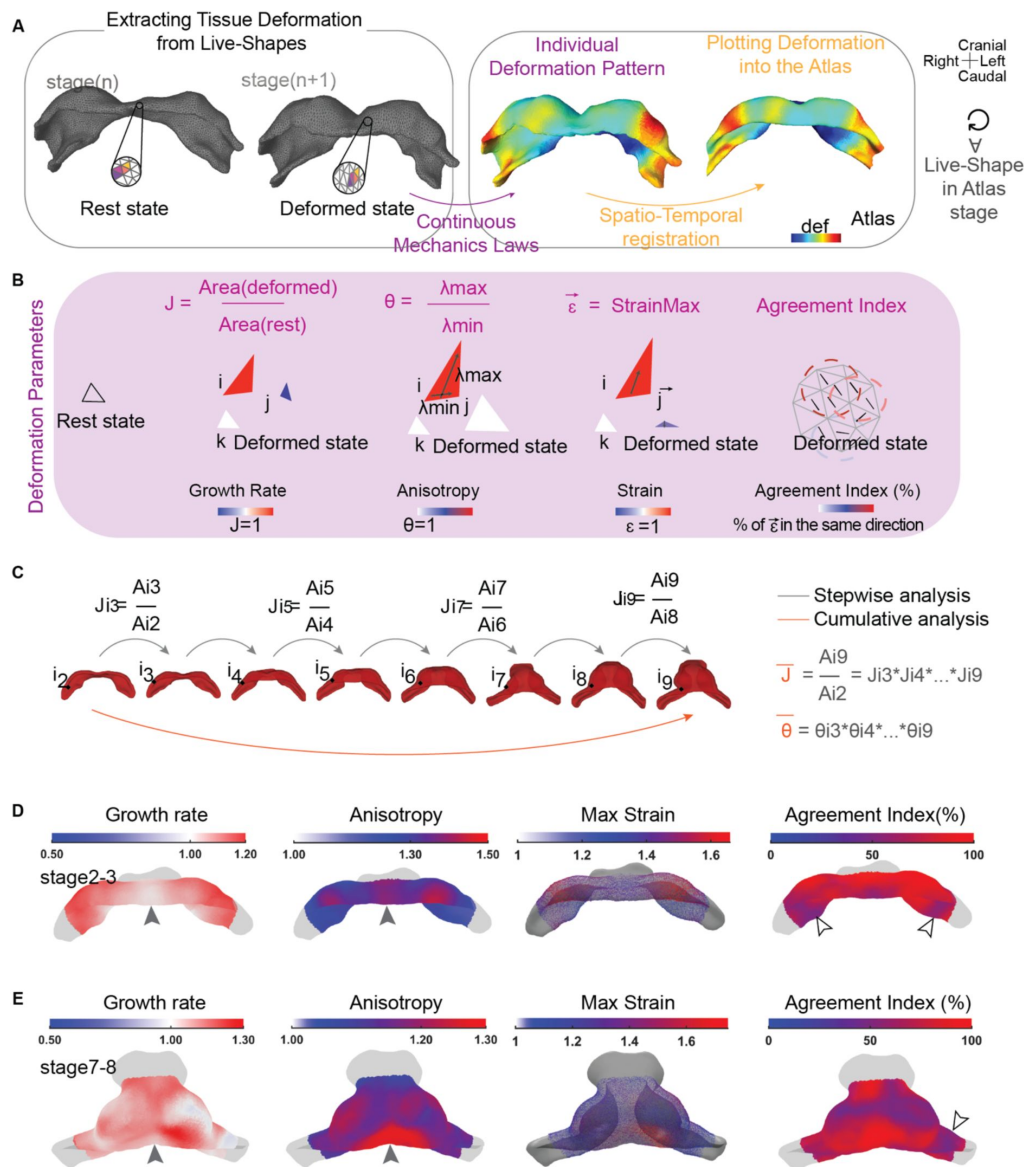


Figure 2

Deformation pattern analysis.

A), Mesh transformation from the initial state to the deformed state. The triangles of the mesh are transformed according to the MIRT tensor from time t_i to time t_{i+1} . The deformation pattern is calculated by applying the principles of continuum mechanics between the initial and deformed states. This pattern is then visualized on the deformed shape. The spatio-temporal registration onto the Atlas enables the mapping of the deformation into it. B), Description of the deformation parameters between the initial and deformed states. For each triangle, three different possible deformations (a, b, c) are represented, color-coded according to the parameter value. The growth rate (J) is defined as the ratio of the area of the same triangle before and after deformation. Anisotropy (θ) is defined as the ratio between the maximum and minimum deformation magnitudes. The maximum strain (ϵ) is characterized by the vector of maximum deformation. The agreement index reports the local coherence in the direction of tissue deformation. C), Difference between stepwise deformation pattern analysis (grey arrows) and cumulative deformation pattern analysis (orange arrows). D-E), Stepwise deformation pattern analysis in caudal view. The color maps represent the growth rate, anisotropy, strain magnitude and direction, and finally, their agreement. Here, the deformation maps are shown for the transition from stage 2 to stage 3 and from stage 7 to stage 8. The full arrowhead indicates a zone of low growth rate and high anisotropy in the most ventral-medial region. The empty arrowhead indicates discontinuity in the agreement of the strain direction. Deformations values are the mean values. The number of specimens averaged for each stage is reported in [Figure 4C](#) in Raiola et al., 2025b.

morphogenesis. This compartmentalization affects both the anisotropy/growth patterns and the directionality of tissue deformation. Furthermore, this analysis detected a strong anticorrelation between tissue growth and anisotropic deformation.

In stage 5 and stage 6, the medio-caudal myocardium continued to deform anisotropically along the craniocaudal direction, with a high rate of agreement, indicating coordinated deformation (**Figure S2** [↗](#)). At this stage, high growth rates localized to the ventrolateral regions and extend into the inflow regions from the lateral regions of the forming ventricle (**Figure S2** [↗](#)). In general, high growth rates are mostly present in the caudal regions of the myocardium, except for some medio-lateral areas of the forming ventricle at stage 6 (**Figure S2** [↗](#)). This is accompanied by the appearance of regions of low agreement in the bilateral bulging areas of the forming ventricle (**Figure S2** [↗](#)). With some exceptions, growth and anisotropy continued to show anti-correlated patterns (**Figure S2** [↗](#)). The inflow regions again showed sharp boundaries between regions of high agreement with discrepant directions, indicating high compartmentalization of the deformation directions.

In stage 7 and stage 8, the caudal region of the myocardium showed reduced growth, whereas new high-growth areas appeared in the ventromedial region of the forming ventricle, extending bilaterally towards the cranial region around the OFT (**Figure 2E** [↗](#), **Figure S3** [↗](#)). In all regions of the forming ventricle, growth correlated with a high index of agreement and, in general, with low anisotropy (**Figure S3** [↗](#)). The inflows continued showing frontiers of strong disagreement separating the distal inflow, predominantly showing a medio-lateral deformation direction, from more proximal regions, showing craniocaudal deformation directions (**Figure S3** [↗](#)). In stage 8, a left-side-specific increase in growth was observed in the ventral side of the ventricle.

In stage 9, the most prominent new feature was the rightward rotation of the ventricle (**Figure S3** [↗](#)), a phenomenon related to heart looping (Le Garrec et al., 2017 [↗](#)). This was accompanied by a sharp increase in growth and decrease in anisotropy at the left junction between the IFT and the ventricle, whereas the opposite took place on the right side (**Figure S3** [↗](#)). As in stage 8, higher growth was also observed in the ventral left side of the ventricle, suggesting its relationship to ventricle rotation.

An interesting observation throughout this study is the behavior of the myocardial rim in contact with the splanchnic mesoderm at the prospective dorsal pericardial wall. This border of the CC deforms drastically towards the cranial pole to form the rim of the OFT and the dorsal myocardial lips that fuse forming the mesocardium. This complex deformation is essential for the transformation of the CC into the primitive HT. The dorsal and ventral views of the myocardium from stage 3 to stage 6 (**Figures S1** [↗](#), **S2** [↗](#)) show that this rim is divided into two regions with different behaviors; a medial region that coincides with the closing dorsal lips and deforms towards the cranial pole, and bilateral distal regions, remaining at their original position. Interestingly, the medial aspects of this rim show very reduced growth or even contraction, whereas its lateral aspects show high growth rates. These results thus show again a strong compartmentalization of growth and deformation during HT formation.

Cumulative Deformation Pattern Analysis

To determine the accumulated changes through several stages, we implemented an additional strategy to analyze tissue deformation in chosen time windows (Raiola et al., 2025b [↗](#)). The purpose was to recreate the continuous motion profile of the heart tissue during its transformation from CC to HT, exploiting the extracted kinetic information from individual live-imaging sequences. The implemented strategy concatenates the motion profiles derived from the live images to fully cover from stage 2 to 9, allowing the cumulative deformation analysis (Raiola et al., 2025b [↗](#)) (**Figure 3A** [↗](#)).

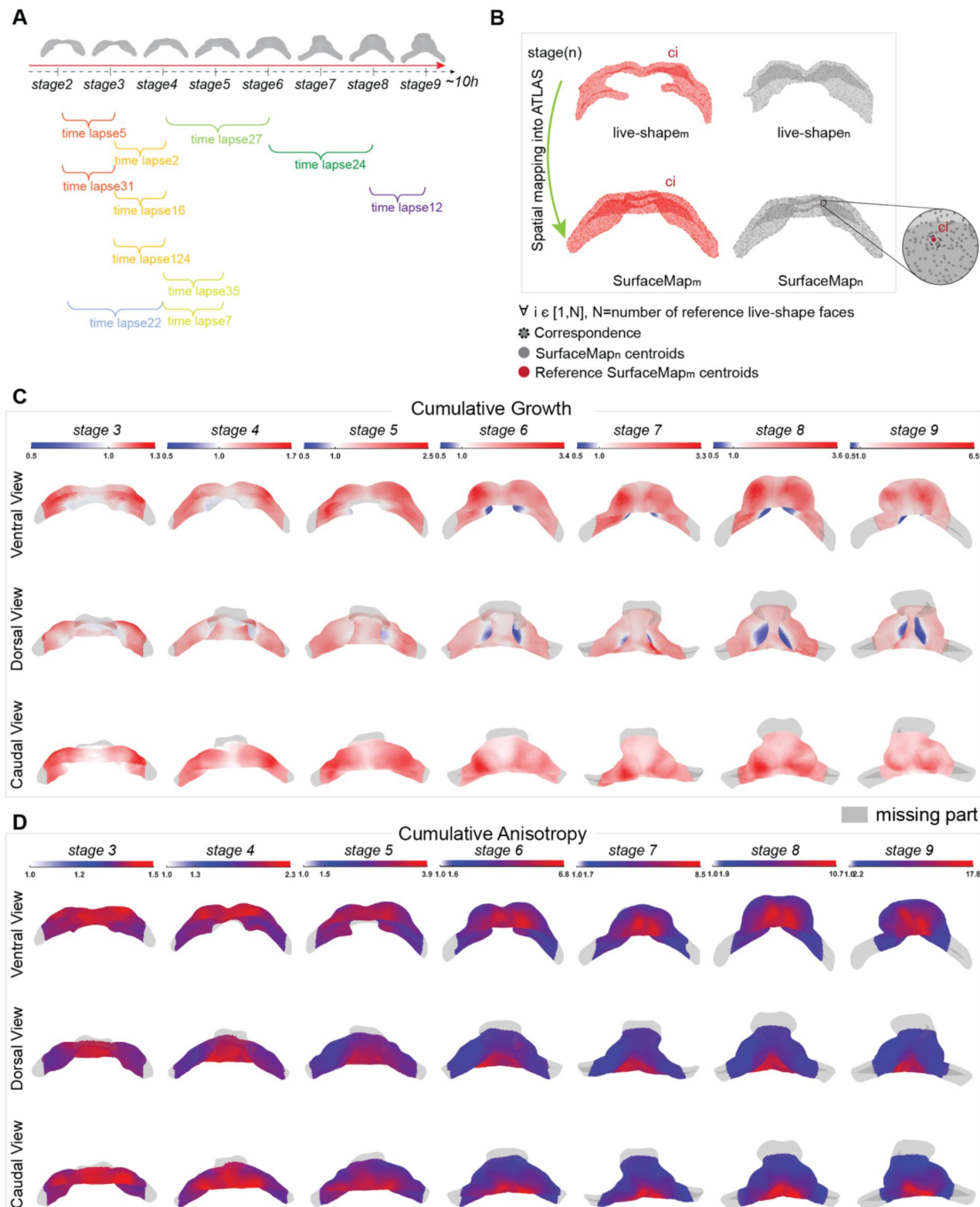


Figure 3

Cumulative Deformation Pattern Analysis.

A), Concatenation of multiple time-lapses on a common timeline. Each time-lapse covers sub-windows of the Atlas temporal line. B), Schematic description of the concatenation pipeline. We fix a reference sample and for each equally staged sample, we select the centroids of the SurfaceMap closest to the reference SurfaceMap. C), Cumulative growth of HT. The color map refers to the mean growth rate of the tissue from stage 2. D), Cumulative anisotropy of HT. The color pattern indicates which zone of the myocardium accumulates more anisotropic deformation. The color map refers to the mean accumulated anisotropy of the tissue from stage 2. The color bars indicate the deformation magnitude. Grey zones correspond to the missing IFTs and OFT parts. Deformations values are the mean values. The number of specimens averaged for each stage is reported in **Figure 4C** in Raiola et al., 2025b.

We used the projections of the Live-Shapes onto the Atlas, which we called “SurfaceMap” (**Figure 3B** [↗](#)), to establish correspondence between specimens and concatenate their motion profiles (Raiola et al., 2025b [↗](#)). In this way, we could map the accumulated growth from stage 2 through the different stages. We found that the highest growth accumulated in the IFTs and at bilateral ventrolateral areas of the forming ventricle, whereas the midline and the OFT remained at low growth rates (**Figure 3C** [↗](#)). The two ventricular bilateral growth regions appeared as focal areas, with a clear single center from which growth rates decline concentrically (**Figure 3C** [↗](#), ventral view). These high-growth foci progressively displace medially during development but always keeping bilaterality up to stage 9. The rim of the myocardium in contact with the prospective dorsal pericardium again showed two distinct regions; a bilateral distal region of relatively high growth in the IFTs, and the region of the dorsal closure, which shows low growth (**Figure 3C** [↗](#)). In contrast, anisotropic deformation accumulated along the ventral midline and the dorsal closure region of the myocardial rim, whereas it remained low in the expanding regions of the tissue (**Figure 3D** [↗](#)). These results confirm the strong compartmentalization of growth and deformation, with growth predominating in lateral regions, and deformation along the craniocaudal direction in medial regions.

In-silico Fate Map Analysis and Its Use in Determining Tissue Motion during Heart Tube Morphogenesis

Next, we used the concatenation strategy to create an *in-silico* fate map reproducing the motion profile of the myocardium during early morphogenesis. This strategy involved displacing the reference SurfaceMap nodes in accordance with the local motion profile, generating a “Dynamic Atlas” (Raiola et al., 2025b [↗](#)). In this Dynamic Atlas, every vertex of the model can be tracked from stage 2 to stage 9 (see Video S1 [↗](#)). Given the correspondence between the model and the actual cell displacement (Raiola et al., 2025b [↗](#)), the resulting *in-silico* fate map can be interpreted in terms of pseudo-cellular displacement within the myocardium. We next validated the Dynamic Atlas by developing the tool “fate map”, which allows to label any position/s and determine its fate *in-silico*. At the regional level, we compared the predicted regional growth in the Dynamic Atlas with that directly measured from the *ex-vivo* labeled cultured embryo. We performed experiments on *ex-vivo* cultured embryos using the TAT-Cre injection (Sendra et al., 2023 [↗](#)) and dye labelling (Domínguez et al., 2012 [↗](#)) to experimentally determine the fate of specific CC regions (**Table S1** [↗](#)). Using the TAT-Cre technique, we labelled a discrete group of cells in the medial CC and tracked the positions of the cell progeny after culturing the embryo for 20 hours (**Figure 4A** [↗](#)). An equivalent label generated in the stage 2 virtual model generated a labelled region in the stage 8 virtual model similar to that obtained in the experimental setting (**Figure 4A, A'** [↗](#)). Next, we labelled a group of embryonic cells with a dye and examined the fate of the label after 15 hours (**Figure 4B** [↗](#)). We then labelled equivalent positions in the stage 2 virtual model and found that the fate of this region in the stage 6 virtual model was similar to that obtained in the experimental setting (**Figure 4B, B'** [↗](#)).

Once determined that the virtual model approximates the behavior of myocardial cells during HT formation, we used it to map the primordia of the different regions of the HT observable at stage 8 (**Figure 4C-F** [↗](#)). We labelled the OFT, the left ventricle primordium (outer curvature), the ventricular side of the future inner curvature (dorsal side of the HT opposing the forming ventricle), and the IFTs. The comparison between the primordia and the end-stage structures indicates that the lateral-most parts of the CC grow and extensively migrate medially to generate the IFTs and the dorsal part of the HT (future inner curvature), whereas the primordia of the future ventricle (outer curvature) and the OFT mainly undergo deformation to achieve their final shape. Whereas both the OFT and the future ventricle reshape towards the midline, the OFT region contracts to a cylindrical shape, while the future ventricle bulges out in a barrel shape (**Figure 4C-F** [↗](#)).

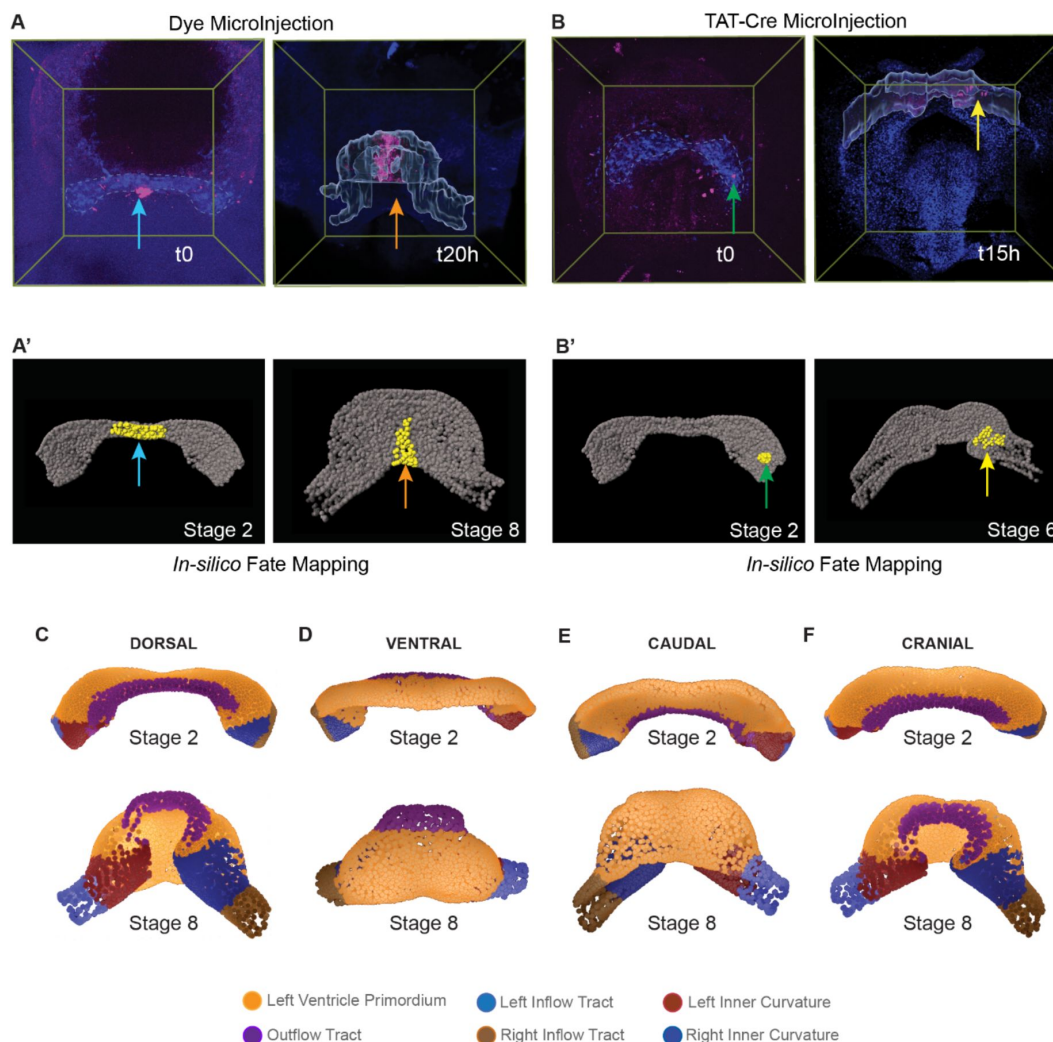


Figure 4

In-silico Fate Map.

A), Labelling of a cardiac crescent discrete region (t0, left) by TAT-Cre- induced recombination and 3D reconstruction of its contribution to the HT after 20 hours of embryo culture (t20, right). A'), labelling on stage 2 virtual model of an equivalent region to that experimentally labelled in A) (left) and the resulting labelled region in stage 8 virtual model (right). B), Labelling of a CC discrete region by dye injection (t0, left) and 3D reconstruction of its contribution to the HT after 15 hours of embryo culture (t15h, right). B'), labelling on stage 2 virtual model of an equivalent region to that experimentally labelled in A) (left) and the resulting labelled region in the stage 6 virtual model (right). C-F), different views of the virtual model showing different regions of the HT in stage 8 and their primordia in stage 2.

Next, to understand how the ventricular barrel shape is produced, we used the virtual model to label groups of cells at regular spacing along the rims between the splanchnic mesoderm and the CC at stage 2 (**Figure 5**). These two rims define two lines, one between the CC and the juxta-cardiac field (**Figure 5A**; D2, as described in [Esteban et al., 2022](#)) and another between the CC and the SHF (**Figure 5B**; D1, as described in [Esteban et al., 2022](#)).

This study shows that along D2 only the lateral-most sections expand towards the midline to generate the sections of D2 underlying the IFTs at stage 8 (**Figure 5A, A'**). In contrast all D2 sections underlying the ventricle at stage 8 undergo a strong compression from their initial length at stage 2 (**Figure 5A, A'**). Whereas the labels that end up positioned within the IFTs show limited cranial expansion, those underlying the ventricle extensively expand cranially and colonize the caudal part of the ventricle (**Figure 5A, A'**). At stage 8, the future left ventricle shows two hemi bulges with an indentation between them corresponding to the midline. The strongest lateral compression and cranial expansion are found for bilateral groups of cells close to the midline of these two hemi bulges, while the midline label expands only moderately towards the ventricular wall (**Figure 5A, A'**). Interestingly, the cranial extensions of these bilateral cell groups open laterally as they extend cranially, generating a fan-like label (**Figure 5A, A'**). These observations suggest that the lateral constriction and cranial expansion of cells close to D2 contribute to generating the barrel shape of the ventricle.

The same study on D1 shows that the region fated to the OFT compresses to form the cylindrical shape of this structure, whereas the lateral aspects of D1 expand drastically to form the cranial border of the IFTs and the dorsal borders of the ventricular region, which will later fuse to form the mesocardium (**Figure 5B, B'**). In this future mesocardial region, cells originally positioned close to D1 expand to populate the inner curvature of the ventricular region (**Figure 5B, B'**). These deformations ensure that the HT forms by generating the inner curvature regions and allowing its dorsal closure. At the same time, the constriction imposed by the formation of the OFT limits the expansion of the ventricle at its cranial end, thereby contributing to its barrel shape.

Finally, to understand the shape changes taking place during the formation of the primitive left ventricle, we labelled longitudinal rectangles in the ventricle of the stage-8 model and studied the evolution of their shape from stage 2 to stage 8 (**Figure 6A**). The observations indicate that the whole ventricular primordium undergoes extensive reshaping with progressive constriction of the tissue towards the midline in a craniocaudal gradient. The movement resembles the closing of an open fan with the vertex at the cranial side (**Figure 6B**). Through this movement, lateral tissue is recruited towards the midline and extends cranial ward. An exception to this general reshaping is the mid-central region of the ventricle, where bilateral deformation takes place in mirror-image directions with an equatorial axis of symmetry. This deformation pattern correlates with the localized fan-like distribution of the descendant of D2 (**Figure 5A, A'**, indicated by asterisks in **Figure 6**). These observations therefore match the deformations observed for labels in D1 and D2 (**Figure 5**). Together, these results show that the ventricle primordium is narrower at the cranial end of the CC and extends progressively more laterally towards the caudal end of the CC. Ventricle morphogenesis thus involves tissue constriction by two “belts” of tissue that support the barrel shape of the ventricle. The cranial belt constitutes the OFT and the caudal belt coincides with the myocardial region abutting the juxta-cardiac field. This caudal belt undergoes a much more extensive constriction than the cranial one, related to the recruitment of very lateral aspects of the CC. In contrast, at the cranial aspect of the CC, this constriction is less strong and ventricle formation does not involve the recruitment of the lateral-most CC regions.

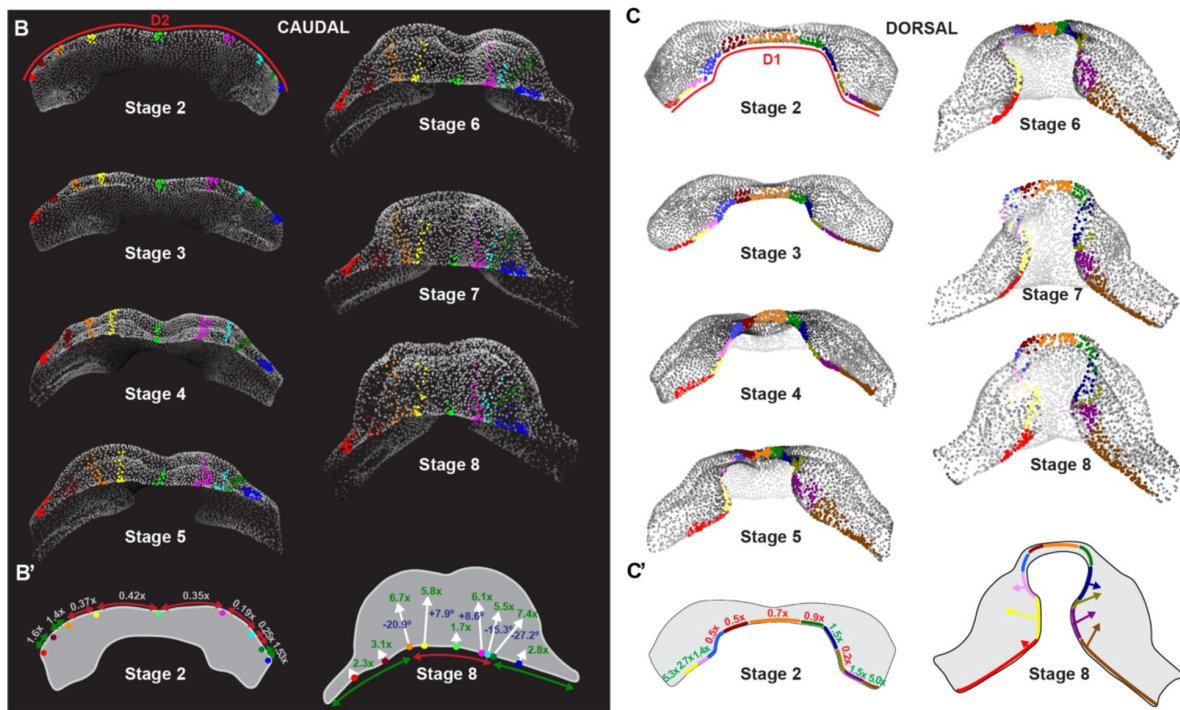


Figure 5.

Fate maps of the cardiac crescent boundaries reveal tissue dynamics of heart tube formation.

A), Clusters of pseudo-cells were labelled at regular spacing along the D2 line of the stage 2 CC model. The labeled clusters were then followed through stage 8. Caudal views are shown, which allow visualize the caudal side of the developing inflows and ventricle. A') Left, representation of the degree of expansion (green) or contraction (red) of the different D2 segments defined between the labelled clusters at stage 2. Numbers indicate fold change from the initial to the final time points. Right, representation of the stage 8 view indicating the regions of D2 that contract (red) versus those that expand (green). Fold-change cranial ward expansion of each labelled cluster from their original craniocaudal extension is shown in green. The direction of the white arrows indicates the main direction of expansion. The angle of the main expansion direction with respect to the embryo mid-plane is indicated in blue. B), Contiguous segments of pseudo-cells were labelled at regular spacing along the D1 line of the stage 2 CC model. The labeled clusters were then followed through stage 8. Dorsal views are shown. B') Left, representation of the labelled segments on stage 2 CC with indication of their fold-change expansion (green) or contraction (red). Right, Representation of the final extension at stage 9 of the D1 labelled segments. Arrows indicate the direction and extension of the expansion of the labelled pseudo-cells to colonize the dorsal parts of the HT.

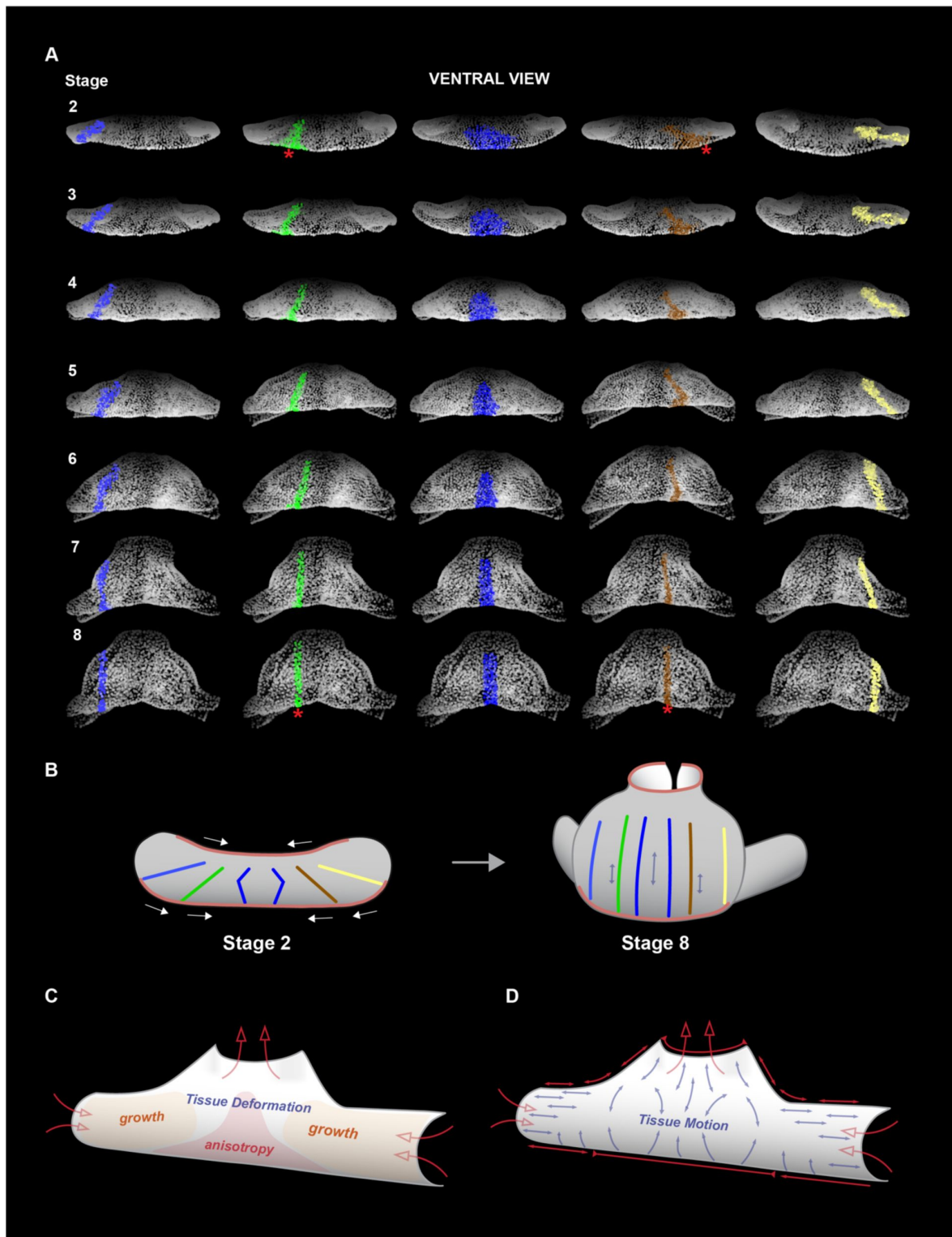


Figure 6.

Fate map of the ventricle primordium shows the tissue dynamics underlying ventricle formation.

A), longitudinal rectangular groups of pseudo-cells were regularly labeled in the ventricle of the stage 8 model. The labelled regions were then tracked back through stage 2. B), Schematic representation of regional deformations at stage 2 and stage 8. C) Representation of the main growth and anisotropy areas in the myocardium. D) Representation of the main directions of tissue motion during heart tube formation.

Discussion

Here we used an image-based pipeline to extract and compare myocardial deformation during HT morphogenesis in the mouse embryo. Our analysis comprises 16 specimens that have been staged into 9 different canonical shapes at nominal ages ranging from E7.75 to E8.25 (approximately 12h). The canonical shapes provide a spatiotemporal reference for comparing and compiling the motion and deformation of multiple specimens.

The knowledge of the deformation patterns extracted from the videos allowed the cumulative evaluation of the most important transformations of the myocardium, measuring their intensity and direction, and determining their timing. The maps reveal a strong compartmentalization of growth rates and anisotropic deformation. Growth is generally high in the lateral regions of the forming ventricle, with a bias towards its caudal parts. In contrast, the medial stripe of the CC – later of the forming tube – does not appreciably grow but shows high anisotropic deformation with stretching mainly oriented in the rostro-caudal direction. Similar mutual exclusion of the high-anisotropic areas and the growing bilateral caudal regions is clearly maintained until stage 6. This compartmentalization generates clear boundaries between regions that show coordinated stretch directions and those that do not and between regions showing different stretching directions (**Figure 6C** [↗](#)). These boundaries are particularly pronounced in the inflow region, where differences in stretching directions become particularly evident (**Figure 6D** [↗](#)). In the IFT, boundary regions suggest two forces that act in different directions: one, stretching the IFT cells in the latero-medial direction, and another promoting the elongation of the tube in a craniocaudal direction. On the dorsal side of the forming tube, the rim that connects the myocardium to the splanchnic mesoderm (D1) also shows a high mediolateral compartmentalization, with its medial part undergoing little growth or even compression at several stages, whereas its lateral parts grow and stretch toward the midline (**Figure 6D** [↗](#)). This originates a progressive cranial looping of the medial aspects of this rim, which form the OFT and the closing mesocardium, whereas the lateral-most parts remain as IFTs. Given that tissue compression inhibits proliferation through the activation of the Hippo pathway (Von Gise et al., 2012 [↗](#)), we propose that one of the factors contributing to the lower growth of the medial regions might be tissue compression. Alternatively, or in addition, it is possible that forces arise from the splanchnic mesoderm, which is highly proliferative (de Boer et al., 2012 [↗](#)) and could contribute to the anisotropic deformation of the mesocardial rim of the myocardium.

Based on these findings, we propose a model for mouse primitive HT morphogenesis in which areas of growth are predominantly lateral in the CC and later become ventrocaudal during the formation of the ventricle. Given that the lateral extension of the heart remains stable during HT formation (Esteban et al., 2022 [↗](#)), we propose that the growth of these lateral regions causes the medial convergence of the tissue. This convergence is driven by two belts that constrain the CC towards the midline, a cranial one that coincides with the OFT and a caudal one at the boundary between the ventricle primordium and the juxta-cardiac field (**Figure 6D** [↗](#)). In between these two belt-like constraints, the ventricle adopts a bulging barrel shape (**Figure 6B** [↗](#)). Surprisingly, ventricle morphogenesis at this stage does not involve localized areas of growth but rather, extensive remodeling to recruit lateral tissue and relocate it along the craniocaudal direction (**Figure 6B** [↗](#)). The model thus proposes complementary regions in which growth and deformation are anticorrelated. Concomitant with this recruitment to medial regions, HT tissue is deformed in the craniocaudal directions, so that the cranial and caudal belts move apart from each other and the HT adopts its longitudinal disposition (**Figure 6B** [↗](#)).

Between stage 7 and stage 9, we observed a difference in growth rate between the two sides of the forming ventricle, consistent with de Boer et al., 2012 [↗](#). This bias could be the first input for the looping process. Increased growth in the left region, as observed in the chicken model, could

influence the direction of cardiac looping and result in a rightward C-shape (Kidokoro et al., 2008). As a consequence of the slight bending, the tissue could undergo compression on the right side. Alternatively, the observed bias could derive from differential growth of myocardial cells during looping (Ebrahimi et al., 2022; Shi et al., 2014). In addition, differential L-R proliferation in the dorsal pericardial wall may also influence heart looping by influencing the dorsal myocardial rim (Desgrange et al., 2020).

By concatenating the motion profiles of multiple specimens, we also presented the first *in-silico* Dynamic Atlas of HT formation, which allows virtual fate mapping experiments. This model shows the strong anisotropic deformation of the tissue, especially in the ventricle and in the dorsal myocardial boundary, where the cumulative deformation analysis suggests the greatest anisotropy. Although the accuracy of the kinetic motion of the fate map has not been directly demonstrated, we believe that the fate map can correctly predict the cardiomyocyte motion in that area, as tested by comparison with the video-microscopy dataset (Raiola et al., 2025b). We propose the Dynamic Atlas and its tool “fate map” as an important approach to explore HT morphogenesis.

In conclusion, the proposed workflow provided detailed kinetics of the myocardium and the deformation underlying early development of the mammalian heart. The fate map is proposed as an innovative tool to shed light on regionalized cell coordination, generating new questions about the biological and genetic factors that regulate the cardiac development process.

Materials and methods

Mouse Strains

Animals were handled in accordance with CNIC Ethics Committee, Spanish laws and the EU Directive 2010/63/EU for the use of animals in research. All mouse experiments were approved by the CNIC and Universidad Autónoma de Madrid Committees for “Ética y Bienestar Animal” and the area of “Protección Animal” of the Community of Madrid with reference PROEX 220/15. For this study, mice were maintained on mixed C57Bl/6 or CD1 background. We used the following mouse lines, which were genotyped by PCR following the original study protocols. Male and female mice of more than 8 weeks of age were used for mating.

Cell Tracking with TAT-Cre Microinjection and Dye Microinjection

Mouse embryos at developmental stages E6.5 to E7.5 were dissected in a pre-equilibrated medium containing DMEM supplemented with 10% fetal bovine serum, 25 mM HEPESNaOH (pH 7.2), penicillin, and streptomycin. Subsequently, the embryos were cultured under controlled conditions within a hypoxic chamber incubator at 37°C with 5% O₂ and 7% CO₂, using a culture medium comprising 50% Janvier Labs Rat Serum SPRAGUE DAWLEY RjHan SD male only and 50% DMEM FluoroBrite. Embryos in the same developmental range were microinjected with TAT-Cre recombinase using specialized equipment and techniques, as previously described (Sendra et al., 2023). Specifically, microinjection needles were prepared with a 2µm gauge and inserted into the anterior side of the embryo until penetrating the endodermal layer, using specified pressure conditions. The embryos were handled and positioned carefully, ensuring that the anterior and posterior sides were oriented accordingly during the procedure to achieve successful microinjections.

We utilised mouse embryos that carried both the reporter genes ROSA26CAG–TdTomato (R26RtdTomato) and ROSA26CAG–EGFP (R26REGFP) in transheterozygosis. We fixed, imaged, and annotated fluorescent cells within anatomical regions. “Clusters” were defined as groups of cells

(either Tomato or GFP) originating from a single TAT-Cre injection. For dye microinjections, mouse embryos were labeled by injection of a lipophilic carbocyanine, DiI (1,1'-dioctadecyl-3,3,3',3'-tetramethylindocarbocyanine perchlorate) as described previously (Franco et al., 2001 [↗](#))

Computational Workflow

The computational workflow is detailed in [Raiola et al. 2025b \[↗\]\(#\)](#).

Strain Agreement Index

We defined the stretch direction as the eigenvector related to the maximum eigenvalue. To better understand the stretching directions at the tissue level, we introduced an additional parameter called the Strain Agreement Index (ϕ). This parameter identified regions of the heart that stretched in the same direction, discriminating them from those with chaotic or discordant directions. ϕ was calculated as the percentage of directions that were in concordance in a certain neighbourhood. We evaluated the ϕ index for each face considering the stretching in a 20-pixel neighbourhood. The chosen neighbourhood is small enough to approximate a flat surface (approximately 6-7 cells). As first step, we projected the selected stretch vectors onto a plane identified by the centroids of the selected faces. For this, we used the *fitPlane* and *projLineOnPlane* functions of the *geom3d* library (<http://github.com/mattools/matGeom/> [↗](#)). In the second step, we calculated the angle formed by each stretch vector with the remaining vectors. We then evaluated the Agreement Index as the percentage of vectors whose angle diverged from $\pm 10^\circ$. The percentage value between 0 and 100% was converted into a color code and plotted for each mesh.

Data and code availability

The authors declare that all the data supporting the findings of this study are available within the article and its supplemental files or from the corresponding author upon reasonable request. All data presented in this study, including raw and processed data have been deposited in a reserved Mendeley Data server at the following address:

- Mendeley Data: <https://data.mendeley.com/preview/nd3kmj3cnx?a=a9cfaa55-3a74-4494-aeae-6cf0e90e5fa8> [↗](#)
- DOIs are listed in the key resources table.

This paper does not report original code.

In-silico fate map is available in <https://github.com/MorRaiola/BarrelHeartModel.git> [↗](#)

Software

We recommend opening the 3D mesh with the open-source software ‘MeshLab’ (<https://www.meshlab.net/> [↗](#)). In ParaView (<https://www.paraview.org> [↗](#)), you can plot the MaxStrain vectors. Once the .vtk file is imported, the vectors can be visualized using the Glyph representation. It is recommended to display the vectors as lines for better visual interpretation. To view and use the *in-silico* Fate Map, the .ims data needs to be imported into Imaris Viewer (<https://imaris.oxinst.com/imaris-viewer> [↗](#)), the free version of the Imaris visualization software.

Supplemental items

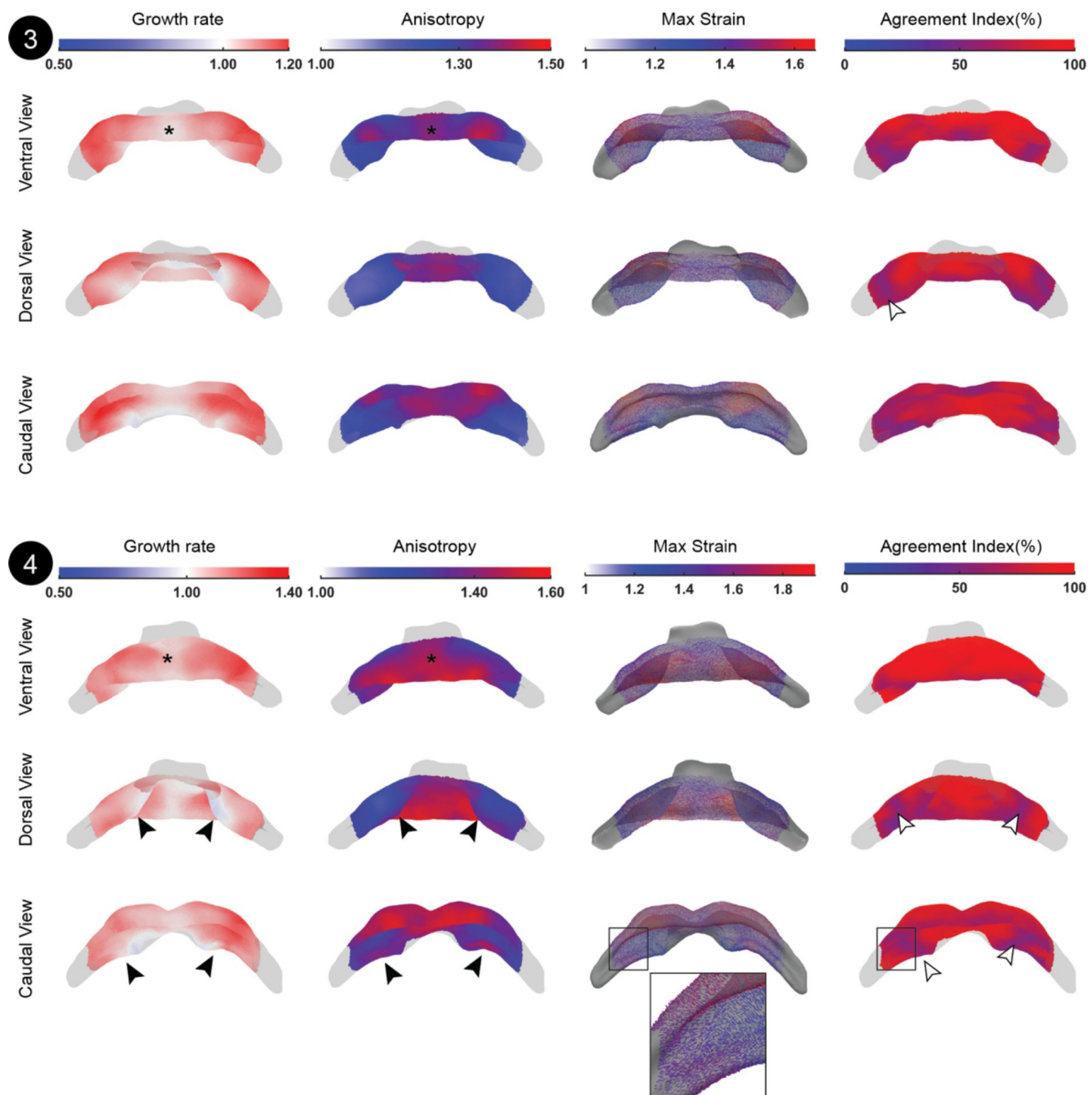


Figure S1

Stepwise Deformation Pattern Analysis.

Deformation analysis for stage 3 and stage 4.

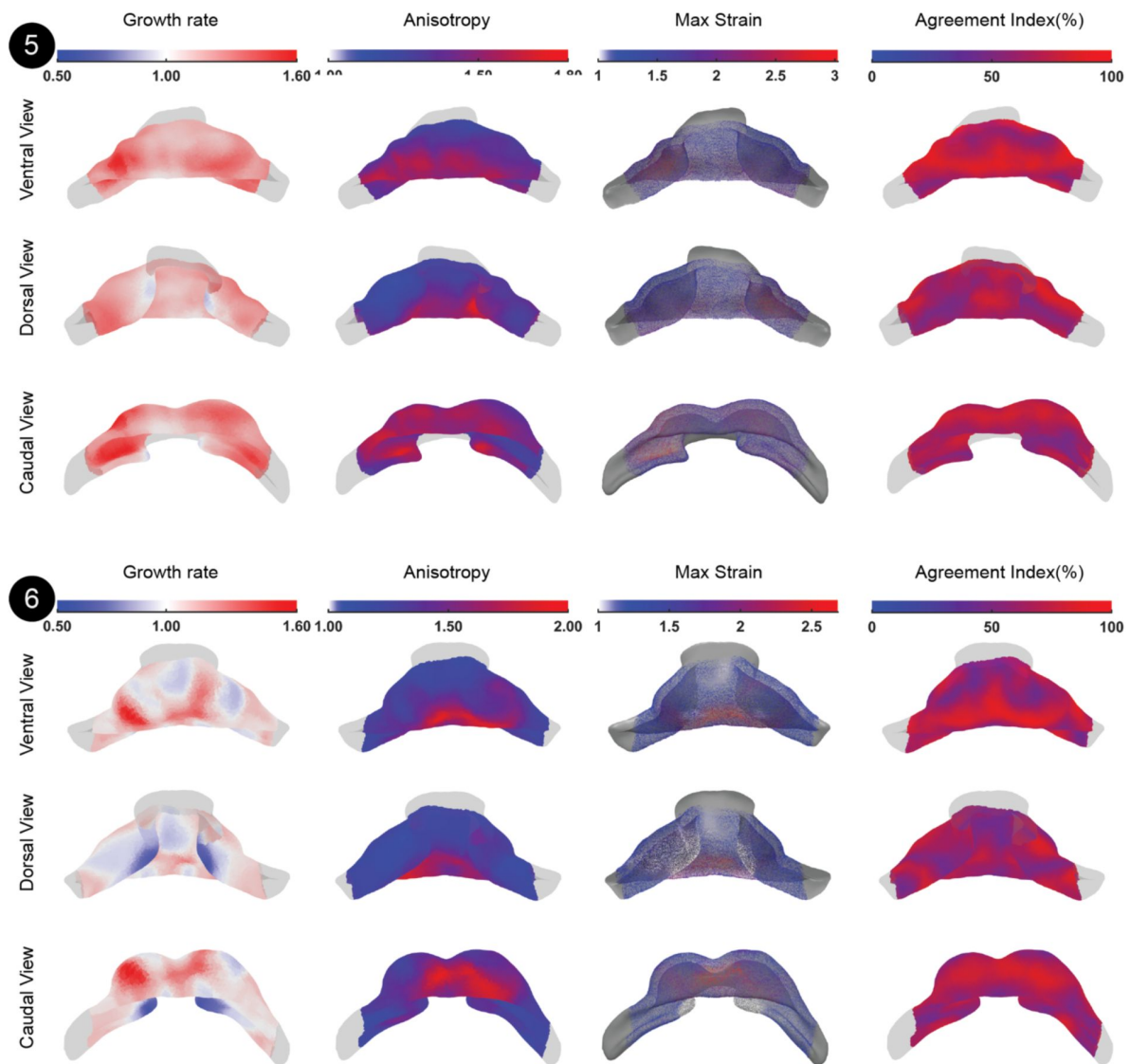


Figure S2

Stepwise Deformation Pattern Analysis.

Deformation analysis for stage 5 and stage 6.

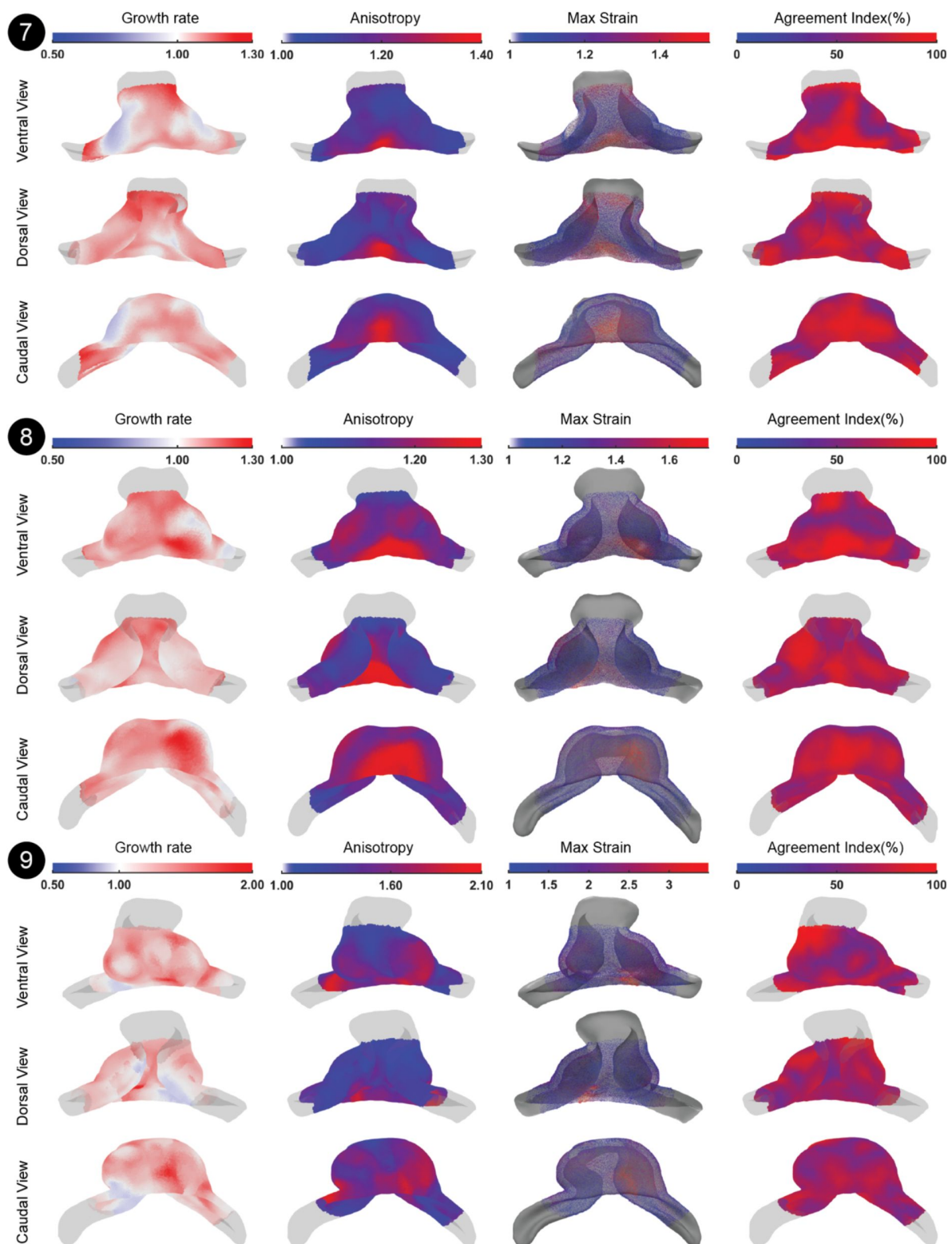


Figure S3

Stepwise Deformation Pattern Analysis.

Deformation analysis for stage 7 and stage 9.

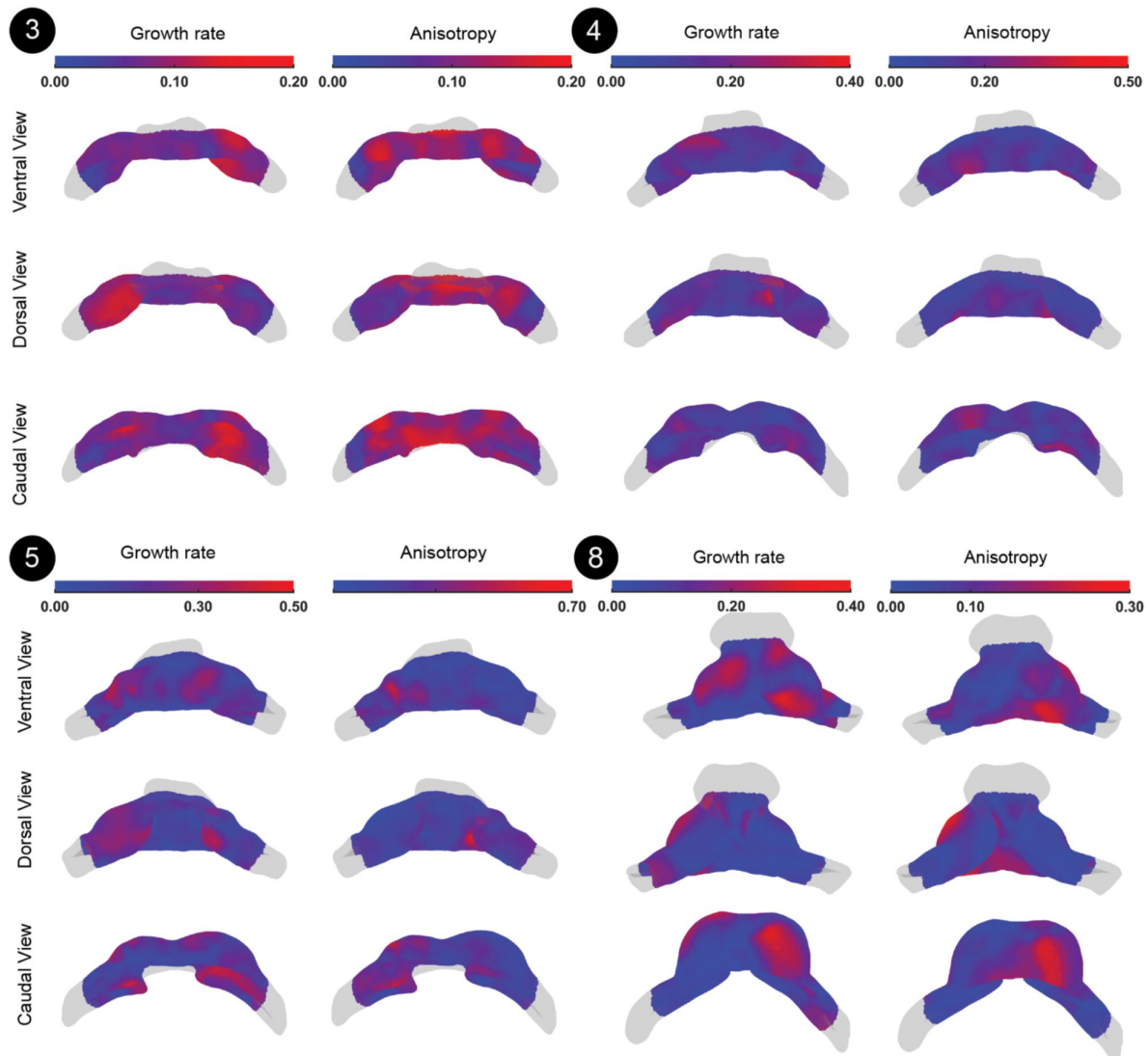


Figure S4

Standard deviation of the growth rate and the anisotropy patterns.

Colormaps indicate the spatial distribution of the Standard Deviations for the growth rate and anisotropy magnitudes shown from Ventral, Dorsal and Caudal views. Stages 6,7,9 have only one staged Live-Shape and therefore variability could not be measured.

Embryo	Genotype	VoxelRes[μm]
eCre(0h)	R26REGFP ; Rosa26Rtdtomato ;	1.136-1.136-2.99
eCre(20h)	Dapi ; R26REGFP ; Rosa26Rtdtomato ; anti-M20(1:100) ;	0.757-0.757-1.04
eDye(0h)	Nkx2.5GFP ; Polr2a–CreERT2 (RERT) ; Rosa26Rtdtomato	1.636-1.636-2.99
eDye(15h)	Nkx2.5GFP ; Polr2a–CreERT2 (RERT) ; Rosa26Rtdtomato	0.734-0.734-6.00

Table S1.

Dataset of TAT-Cre microinjection and cell Dye microinjection.

eCre(20h) image was captured with Flash procedure ([Messal et al.,2021](#) [🔗](#)).

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Additional information

Author contributions

Conceptualization: M.R., M.T.; Methodology: M.S., J.N.D., M.R.; Software: M.R.; Formal analysis: M.R., J.N.D.; Investigation: M.S., J.N.D., M.T.; Data curation: M.R., M.T.; Writing - original draft: M.R.; Writing - review & editing: M.R., M.T.; Supervision: M.T.; Project administration: M.T.; Funding acquisition: M.T.

Additional files

Video S1. *In-silico Fate Map. The Dynamic Atlas is represented in its point-cloud version. At the cellular level, a yellow spot tracks one pseudo cell position throughout heart morphogenesis. At the tissue level, yellow spots illustrate the tissue deformation occurring during heart morphogenesis.* [🔗](#)

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Reviewer #1 (Public review):

Summary:

The study by Raiola et al. conducted a quantitative analysis of tissue deformation during the formation of the primitive heart tube from the cardiac crescent in mouse embryos. Using the tools developed to analyze growth, anisotropy, strain, and cell fate from time-lapse imaging data of mouse embryos, the authors elucidated the compartmentalization of tissue deformation during heart tube formation and ventricular expansion. This paper describes how each region of the cardiac tissue changes to form the heart tube and ventricular chamber, contributing to our understanding of the earliest stages of cardiac development.

Strengths:

In order to understand tissue deformation in cardiac formation, it is commendable that the authors effectively utilized time-lapse imaging data, a data pipeline, and in silico fate mapping.

The study clarifies the compartmentalization of tissue deformation by integrating growth, anisotropy, and strain patterns in each region of the heart.

Weaknesses:

The significance of the compartmentalization of tissue deformation for the heart tube formation remains unclear.

<https://doi.org/10.7554/eLife.108559.1.sa3>

Reviewer #2 (Public review):

The authors address an important challenge in developmental biology: the quantitative description of tissue deformation during organogenesis. They have developed a new pipeline to quantify early heart tube morphogenesis in the mouse, with cellular resolution. They adopt an elegant approach by integrating multiple 3D time-lapse datasets into a dynamic atlas of cardiac morphogenesis in order to compute spatio-temporal deformation patterns. The main findings highlight a strong compartmentalization of cell behaviors, with tissue growth and anisotropy exhibiting complementary and spatially segregated patterns. Using these data, the authors developed an in-silico fate mapping tool to interrogate cell displacement within the myocardium. This virtual model provides new mechanistic insights into how the bilateral cardiac primordia converge and transform into a three-dimensional heart tube. The authors identify "belt-like" constraints at the arterial and venous poles that prevent tissue expansion and thus shape the ventricular barrel morphology.

The computational framework is highly innovative and impressive, providing an unprecedented 3D model of tissue deformation during heart morphogenesis. It also opens avenues for testing hypotheses regarding tissue growth and the forces that cause cell motion. However, the proposed model of ventricular chamber formation with the two constraining belts remains hypothetical, lacking biological validation and requiring strengthening or modulation.

Overall, this carefully performed study provides a new model for exploring tissue deformation during organogenesis and will be of broad interest to computational and developmental biologists.

<https://doi.org/10.7554/eLife.108559.1.sa2>

Reviewer #3 (Public review):**Summary:**

The manuscript by Raiola and colleagues entitled "Quantitative computerized analysis demonstrates strongly compartmentalized tissue deformation patterns underlying mammalian heart tube formation" takes a highly quantitative approach to interrogating the earliest stages of cardiogenesis (12 hours, from early cardiac crescent to early heart tube) in a new and innovative way. The paper presents a new computational framework to help identify both regional and temporal patterns of tissue deformation at cellular resolution. The method is applied to live embryo imaging data (newly generated and from the group's previous pioneering work). In the initial setup, the new model was applied directly to raw time-lapse data, and the results were compared to actual cell tracks identified manually, showing close correlations of the model with the manual tracking. Next, they integrated spatial and temporal information from different embryos to generate a new model for tissue movement, driven by parameters such as tissue growth and anisotropy. Key findings from their model suggest that there are distinct compartments of tissue deformation patterns as the bilateral cardiac crescent develops into the linear heart tube, and that the ventricular chamber forms by a defined expansion pattern, as a 'hemi-barrel shape', with the arterial and venous poles (IFT and OFT) acting as the harnessing belts constraining the expansion of the chamber further. Lastly, the model is tested for its ability to predict future residence of cardiac crescent cells in the heart tube, which it seems to be able to do successfully based on fate tracking validation experiments.

Strengths:

The manuscript provides an exceptionally careful analysis of a critical stage during heart development - that of the earliest stages of morphogenesis, when the heart forms its first tube and chamber structures. While numerous studies have interrogated this stage of heart development, few studies have performed time-lapse imaging, and, to my knowledge, no other report has performed such in in-depth quantitative analysis and modeling of this complex process. The computational model applied to normal heart development of the myocardium (labelled by *Nkx2-5*) has revealed multiple new and interesting concepts, such as the distinct compartments of tissue deformation patterns and the growth trajectories of the emerging ventricle. The fact that the model operates at cellular resolution and over a nearly continuous time period of approximately 12 hours allows for unprecedented depth of the analysis in a largely unbiased manner. Going forward, one can imagine such models revealing additional information on these processes, performing analyses of subpopulations that form the heart, and maybe most importantly, applying the model to various perturbation models (genetic or otherwise). The manuscript is very well written, and the data display is accessible and transparent.

Weaknesses:

No major weaknesses are noted with the study. It would have been very exciting to see the model applied to any kind of perturbation, for example, a left-right defect model, or a model with compromised cardiac progenitor populations. However, the amount of live imaging required for such analyses renders this out of scope for the current study.

<https://doi.org/10.7554/eLife.108559.1.sa1>

Author response:

We are going to modify the text following Reviewer's comments and perform embryo direct labelling experiments to experimentally address the contraction of the two "belts" proposed in our model. We feel that this aspect is feasible in a reasonable time and important for the model proposed. We appreciate the relevance of using this framework to identify molecular drivers of the regionalized tissue behaviours uncovered and how these might be altered in mutant models, but feel that these aspects demand efforts beyond the the reasonable revision periods.

<https://doi.org/10.7554/eLife.108559.1.sa0>